INTRODUCTION TO NETWORKS

Course: Diploma in Cyber Security

Topic: Introduction to Networks

Name: HET ACHARYA

Introduction

A computer network is a group of devices connected together so they can share data and services. We use networks almost every day while using phones, laptops, Wi-Fi, apps and other smart devices.

In this assignment, I learned about different types of networks and how a phone or computer communicates with online services.

TASK 1 – PHONE TO INSTAGRAM CONNECTION

When I open Instagram on my phone, my phone sends the request through the Wi-Fi router. The router connects to the Internet Service Provider (ISP), and the ISP carries the data through the internet to Instagram's servers. The response then comes back through the network to my phone.

Network Flow

PHONE → Wi-Fi → WI-FI ROUTER → ISP → INTERNET → INSTAGRAM SERVER

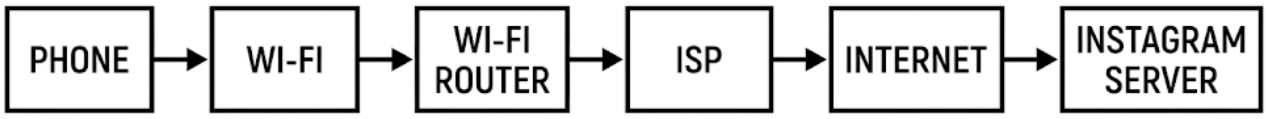


Figure 1: Phone connected to Instagram through Wi-Fi, ISP and Internet

Labels for the diagram

- Phone – My device

- Wi-Fi Router – Connects my phone to the local network
- ISP – Provides the internet connection
- Internet – Connects different networks around the world
- Instagram Server – Receives requests and sends data back

TASK 2 – DAILY NETWORK DEVICES

These are five devices that I commonly use or see around me and the type of network they can connect to.

Device	Network Type	Use
Smartphone	WAN / Wi-Fi	Connects to the internet and online apps.
Laptop	LAN / WAN	Used for browsing, study and sharing data.
Smart TV	LAN / WAN	Connects to Wi-Fi for streaming and online services.
Smartwatch	PAN	Usually connects to a phone using Bluetooth.
Wi-Fi Router	LAN / WAN	Connects local devices and provides internet access.

Note: The network type can depend on how the device is connected. For example, a smartphone can use Wi-Fi on a LAN or mobile data over a WAN.

TASK 3 – INTERNET VS INTRANET

Internet

The Internet is a public worldwide network. People can use it to access websites, apps and online services.

Example: When I use college Wi-Fi to open Instagram or Google, I am using the Internet.

Intranet

An intranet is a private network used inside an organization such as a college or company. It is normally available only to authorized users.

Example: If my college has an internal portal that is available only to students and staff, it can be an example of an intranet.

Main Difference

Internet = Public and worldwide

Intranet = Private and limited to an organization

TASK 4 – HOW DATA TRAVELS IN AN APP

App used: Instagram

When I search for something in Instagram, I type the search word and press the search button. My phone sends the request through Wi-Fi or mobile data to the network and then through the ISP and Internet. The request reaches Instagram's server, where the search is processed. The server sends the results back through the network, and my phone displays the results in the app.

Simple Flow

Phone → Wi-Fi Router / Mobile Network → ISP → Internet → Instagram Server

Instagram Server → Internet → ISP → Router / Mobile Network → Phone

CONCLUSION

In this assignment, I learned the basic idea of computer networks and how different devices communicate with each other.

I understood how my phone connects to an online service such as Instagram using a Wi-Fi router, ISP and the Internet. I also learned about LAN, WAN and PAN with daily-life examples.

I learned the difference between the Internet and an intranet. Finally, I understood that when I search something in an app, my request goes to the app's server and the result comes back to my device through the network.

Overall, this assignment helped me understand networking in a simple and practical way.